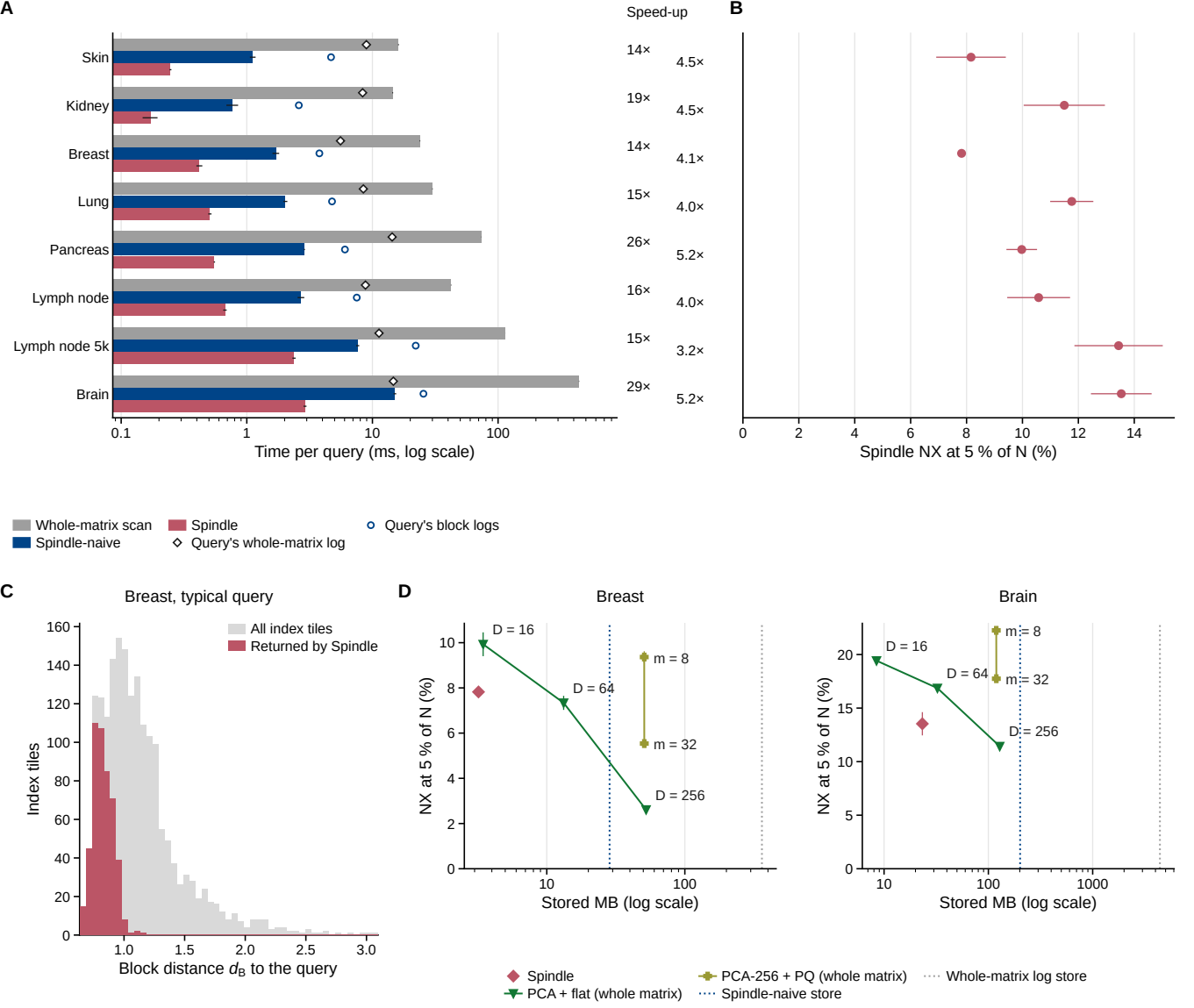
A
Speed-up
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
14×
19×
14×
15×
26×
16×
15×
29×
4.5×
4.5×
4.1×
4.0×
5.2×
4.0×
3.2×
5.2×
0.1
1
10
100
Time per query (ms, log scale)
B
0
2
4
6
8
10
12
14
Spindle NX at 5 % of N (%)
Whole-matrix scan
Spindle-naive
Spindle
Query's whole-matrix log
Query's block logs
C
Breast, typical query
All index tiles
Returned by Spindle
Index tiles
0
20
40
60
80
100
120
140
160
1.0
1.5
2.0
2.5
3.0
Block distance $d_B$ to the query
D
Breast
NX at 5 % of N (%)
D = 16
D = 64
m = 8
m = 32
D = 256
0
2
4
6
8
10
10
100
Stored MB (log scale)
Brain
NX at 5 % of N (%)
D = 16
D = 64
m = 8
m = 32
D = 256
0
5
10
15
20
10
100
1000
Stored MB (log scale)
Spindle
PCA + flat (whole matrix)
PCA-256 + PQ (whole matrix)
Spindle-naive store
Whole-matrix log store